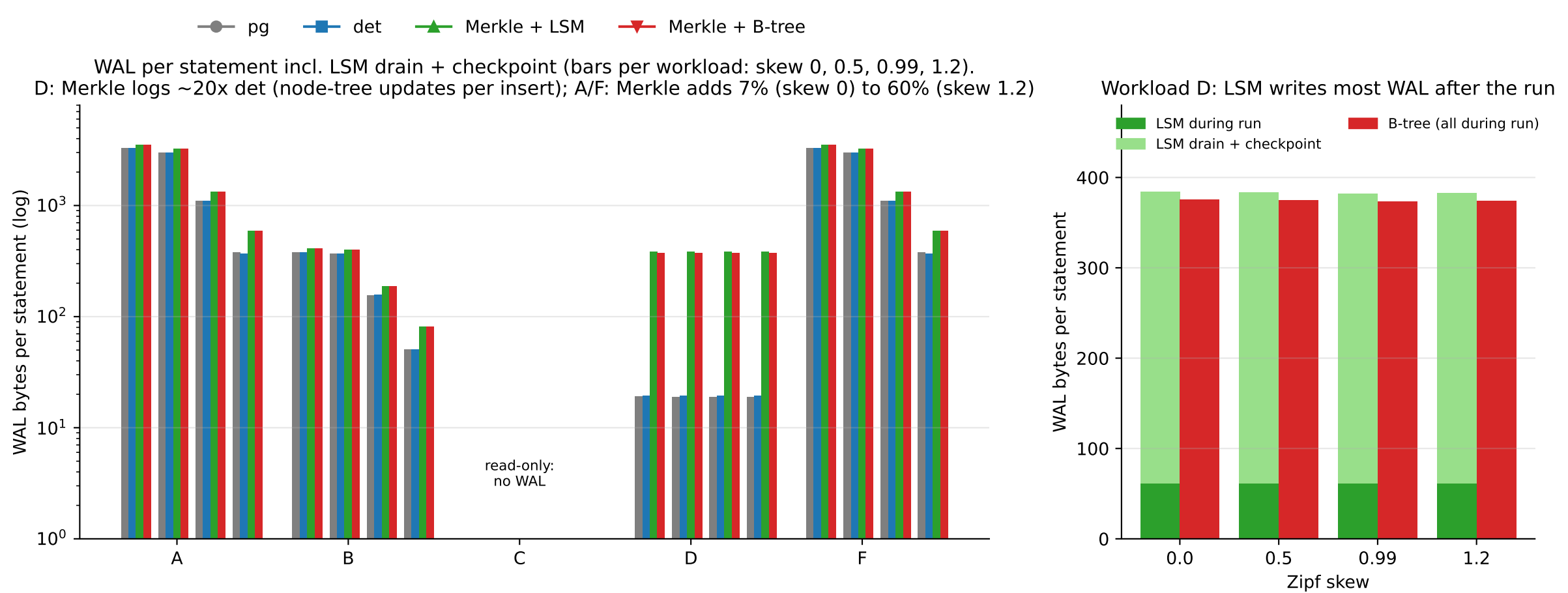
pg
det
Merkle + LSM
Merkle + B-tree
WAL per statement incl. LSM drain + checkpoint (bars per workload: skew 0, 0.5, 0.99, 1.2).
D: Merkle logs ~20x det (node-tree updates per insert); A/F: Merkle adds 7% (skew 0) to 60% (skew 1.2)
WAL bytes per statement (log)
10^3
10^2
10^1
10^0
read-only:
no WAL
A
B
C
D
F
Workload D: LSM writes most WAL after the run
LSM during run
LSM drain + checkpoint
B-tree (all during run)
WAL bytes per statement
400
300
200
100
0
0.0
0.5
0.99
1.2
Zipf skew